



CSE6364 SOFTWARE EVOLUTION & MAINTENANCE
MARCH/APRIL 2026 (TERM 2610)
PROJECT DESCRIPTION

Significant Dates:

- Group formation (Week 02) : 6 Apr 2026 – 12 Apr 2026, 12 midnight
- Submission Project
 - Part I (Proposal) : 26 Apr 2026, 12 midnight (Week 04)
 - Part II (Full report) : 14 June 2026, 12 midnight (Week 12)
- Presentation : 15 – 28 June 2026 (Week 13-14)

* Please strictly adhere to the important dates above.

* * A kind reminder for every students that **penalty for late submission** could be applied.

Instructions:

Students need to form a **group of 3 to 4** from the **SAME TUTORIAL** section. In this group assignment, students will engage with the real-world process of software maintenance and evolution by selecting an open-source software project and contributing meaningful improvements or enhancements. Each group is required to identify an open-source project from reputable platforms such as GitHub or GitLab. The chosen project must have sufficient documentation and a codebase manageable within the assignment's timeline.

Note: Name your file in the following format:

Proj_Px_TTxL_Gx_stud1, stud2, stud3, stud4.pdf

Example:

Proj_Px_TT3L_G5_David, Edwin, Ella, Othman.pdf

Proj: obviously this is Project.

Px: Project part: PI or PII,.

TTxL: your tutorial class code: TT1L, TT2L, TT3L, etc.

Gx: your group number.

stud: group member's name (short one, EX: Studname: **David** Fong Seow Kee)

Project Description:

Student are required to:

- 1) Explore the selected codebase, analyzing its structure, functionality, and areas for potential improvement. Students are encouraged to review documentation, understand the software's purpose, and identify issues such as bugs, performance bottlenecks, or opportunities for feature enhancements.

- 2) Prepare an improvement proposal that outlines the identified problem or enhancement opportunity. This proposal should clearly explain the justification for the proposed change, its anticipated benefits, and a high-level implementation plan.
- 3) Following approval of the proposal, students will implement their changes to the codebase, adhering to software engineering best practices. This includes writing clean, maintainable code and ensuring that all modifications integrate seamlessly with the existing system.
- 4) The changes must be tested thoroughly, validating functionality and demonstrating that their improvements do not introduce new issues. The testing process should be documented and included in the final submission.
- 5) To support the enhancement, students are required to update any relevant project documentation. This may include modifying the README file, providing user guides for new features, or adding inline comments to improve code comprehensibility. A final report summarizing the entire process—from code analysis to implementation and testing—must be submitted. The report should also include challenges faced, solutions implemented, and reflections on the learning experience.
- 6) Lastly, each group will present their work, showcasing their improvements and explaining their impact on the software. This presentation will serve as a platform to demonstrate technical understanding, collaborative efforts, and effective communication skills.

Additional Information.

Associated Documentation for Software Project:

- 1) README File
- 2) Contributing Guidelines
- 3) LICENSE File
- 4) CHANGELOG File
- 5) Installation Guide
- 6) User Guide/Manual
- 7) Developer Documentation
- 8) Test Documentation
- 9) FAQ
- 10) Issue Templates (for GitHub/GitLab)
- 11) Pull Request Template (for GitHub/GitLab)

Proposal Outline:

1. Cover Page
2. Introduction
3. Problem Statement
4. Justification for Improvement
5. Proposed Solution
6. Impact Analysis
7. Timeline
8. Conclusion
9. References (if any)
10. Appendices (if necessary)

Please use the template given and download it from Ebwise.

PROJECT - PART I – PROPOSAL (RUBRIC)- 10%

Criteria	Excellent (Full Marks)	Good (Partial Marks)	Satisfactory (Moderate Marks)	Needs Improvement (Low Marks)	Poor (No Marks)
Clarity and Organization (2%)	Proposal is exceptionally clear, well-structured, and logically organized, making it easy to follow. (2m)	Proposal is clear and mostly well-structured with minor organizational flaws. (1.5m)	Proposal is somewhat clear but lacks logical flow or structure in some sections. (1m)	Proposal is unclear or disorganized, making it difficult to follow. (0.5m)	Proposal lacks clarity and structure entirely. (0m)
Problem Identification (2%)	Clearly identifies a significant and relevant problem, supported by evidence or examples. (2m)	Identifies a relevant problem with adequate detail but lacks strong evidence. (1.5m)	Identifies a basic problem but lacks relevance or sufficient detail. (1m)	Problem identified is vague or of minor relevance to the project. (0.5m)	No clear problem is identified. (0m)
Proposed Solution (3%)	Solution is innovative, highly feasible, and addresses the problem effectively. (3m)	Solution is feasible and adequately addresses the problem with some limitations. (2.5m)	Solution is somewhat feasible but lacks innovation or depth. (2m)	Solution is not feasible or only partially addresses the problem. (1m)	No viable solution is proposed. (0m)
Justification and Impact (2%)	Justification is well-supported, and the proposal clearly outlines significant positive impacts. (2m)	Justification is adequate, with clear but moderate impact on the project. (1.5m)	Justification is weak, and the potential impact is minimal. (1m)	Justification is vague, and the proposal's impact is unclear. (0.5m)	No justification or impact is provided. (0m)
Implementation Plan (1%)	Includes a clear, well-defined, and practical high-level implementation plan. (1m)	Provides a high-level implementation plan with minor gaps in clarity or detail. (0.8m)	Implementation plan is basic and lacks depth or clear steps. (0.6m)	Plan is vague and lacks important details or practicality. (0.4m)	No implementation plan is provided. (0m)

PROJECT - PART II – FINAL REPORT (RUBRIC)-25%

Criteria	Excellent (5 Marks)	Good (4 Marks)	Satisfactory (3 Marks)	Needs Improvement (2 Marks)	Poor (0 Marks)
Report Content and Structure (5%)	The report is comprehensive, well-structured, and covers all required sections with exceptional clarity and detail.	The report is clear, mostly well-structured, and covers most required sections with adequate detail.	The report is somewhat clear but lacks completeness or has significant structural issues.	The report is unclear or incomplete, missing multiple key sections.	The report lacks clarity, structure, and essential content.
Analysis and Implementation (5%)	Implements 4 or more meaningful software enhancements, with detailed analysis and exceptional documentation.	Implements 3 meaningful software enhancements, with good analysis and documentation.	Implements 2 meaningful software enhancements, with basic analysis and adequate documentation.	Implements 1 meaningful software enhancement, with superficial analysis or limited documentation.	No meaningful software enhancements implemented or documented.
Testing and Validation (5%)	Thorough testing is conducted and well-documented, with clear evidence of validation.	Testing is adequately conducted and documented, with minor gaps in evidence or clarity.	Testing is basic and lacks depth or sufficient documentation.	Testing is minimal and poorly documented.	No meaningful testing or validation is provided.
Presentation (5%)	Presentation is clear, engaging, and well-organized, covering all required aspects effectively.	Presentation is clear and mostly engaging, covering most required aspects adequately.	Presentation is somewhat clear but lacks engagement or misses some key aspects.	Presentation is poorly organized, difficult to follow, or misses significant aspects.	Presentation is unclear, disorganized, and fails to cover key aspects.
Peer Review (5%)	Peer reviews indicate excellent collaboration, contribution, and teamwork from all members.	Peer reviews indicate good collaboration and contribution, with minor areas for improvement.	Peer reviews indicate basic collaboration and contribution, with noticeable gaps.	Peer reviews indicate minimal collaboration or uneven contribution.	Peer reviews indicate poor collaboration and lack of contribution.